Ali Ahmadi Esfidi

AI Researcher & Developer

aliahmadiesfidi@outlook.com | Mr-Ahmadi | Ali Ahmadi Esfidi | Portfolio

Summary. I am an AI researcher and developer with a B.Sc. in Computer Science from Amirkabir University of Technology (2026). I focus on deep reinforcement learning and machine learning for real-world decision-making, with published work in network optimization and applied projects in infrastructure monitoring and computational biology. My software and web development background helps me translate research models into practical, reliable tools.

Education

B.Sc. Computer Science | Amirkabir University of Technology — Sep 2022 – Sep 2026

- **Bachelor's thesis:** Machine Learning Perspectives on Gold-Binding Peptides
  - Supervisor: Fatemeh Zare-Mirakabad | Scholar
- Location: Tehran, Iran • Field of study: Computer Science • GPA: 18.93/20 • Top 20%

Research Interests

- Deep Reinforcement Learning
- Machine Learning for Decision-Making Systems
- AI for Infrastructure Monitoring
- Optimization Methods for AI and Networking
- Computational Biology & Bioinformatics
- Software-Defined Networks & Resource Allocation

Publications

Journal Articles

- M. Saberi, A. Ahmadi Esfidi, M. Dolati, N. Omidvar, **FlexINA: In-Network Aggregation for Accelerating Distributed Machine Learning with Flexible Routing**, Published in *IEEE Transactions on Network and Service Management*. View Paper.
- P. Heidari, A. Khonsari, A. Ahmadi Esfidi, A. Mehrvarz, E. Khodaei, A. Dadlani, **DeepIrrigo: Deep Reinforcement Learning for Continuous and Discrete Action Irrigation Optimization — A Case Study in Iran**, Under review at *Intelligent Systems with Applications*. View Paper.
- M. Saberi, A. Ahmadi Esfidi, M. Dolati, A. Khonsari, A. Movaghar, **Rule Caching in Programmable Networks with Deep Reinforcement Learning**, Submitted to *IEEE Transactions on Network and Service Management*.

Conference Papers

- P. Heidari, A. Ahmadi Esfidi, A. Mehrvarz, E. Khodaei, A. Khonsari, **Irrigation Optimization in Agricultural Fields Using Deep Reinforcement Learning Approaches**, Published in *CSICC 2025*. View Paper.

Research Experience

Research Assistant (part-time) | Supervisor: Prof. Ghatee — Jan 2025 – Present

NORC Lab, Amirkabir University of Technology

- **Led** an AI team and **engineered** web systems and deep learning models for automated bridge damage detection in collaboration with the Tehran Urban Planning and Research Center.
- **Gained hands-on experience** in developing efficient deep-learning pipelines for computer vision applications, focusing on model compactness and structural health monitoring.
- *Stack:* `PyTorch` • `OpenCV` • `ReactJS` • `NodeJS` • `PostgreSQL` • `Docker`

Research Assistant (part-time) | Supervisor: Prof. Khonsari — Sep 2024 – Present

High Performance Networks Lab, Tehran University

- **Developed** deep reinforcement learning systems for complex agricultural decision-making, focusing on reward engineering and policy optimization to balance multiple objectives.
- **Acquired proficiency** in benchmarking reinforcement learning algorithms across discrete and continuous action spaces, and in utilizing simulation environments for training and evaluation.
- *Stack:* `PyTorch` • `TensorFlow` • `Pandas` • `NumPy`

Research Assistant (part-time) | Supervisor: Dr. Dolati — Sep 2025 – Present

Sharif University of Technology

- **Formulated** MDPs for switch pairing and collaborative rule caching, and designed simulation environments for joint scheduling-routing policies in programmable data planes.
- **Implemented** optimization frameworks for in-network aggregation in distributed machine learning, focusing on ILP decomposition, flexible routing strategies, and scalable algorithmic design.
- *Stack:* `PyTorch` • `Numba` • `Pandas` • `NumPy`

Teaching Experience

Teaching Assistant (part-time) | Graduate & Undergraduate Courses — Sep 2024 – Present

Amirkabir University of Technology & Tehran University

• **Data Mining** B.Sc.	Dr. Yousefimehr	Sep 2026 – Present
• **Computational Data Mining** M.Sc.	Prof. Ghatee	Sep 2025 – Jan 2026
• **Introduction to Logic** B.Sc.	Dr. Didehvar	Sep 2025 – Jan 2026
• **Theory of Computation** B.Sc.	Dr. Didehvar	Sep 2024 – Jun 2026
• **Artificial Intelligence & Workshop** B.Sc.	Prof. Ghatee, Dr. Yousefimehr	Sep 2024 – Jan 2025

Selected Projects

See more

DRL-Based Beamforming Optimization for Movable Intelligent Surfaces — Jun 2026 – Present

- **Designed** a DRL framework to optimize beamforming for MIS-assisted SWIPT systems, jointly configuring phase shifts and power splitting ratios to maximize energy harvesting.
- *Stack:* `PyTorch` • `Gymnasium` • `MOSEK` • `NumPy`

RL-Based ML Job Scheduler with Distribution Optimization — Aug 2025 – Oct 2025

- **Developed** a two-stage hierarchical PPO system for scheduling ML jobs on multi-server accelerator clusters using the Gavel dataset, reducing average job completion time.
- *Stack:* `PyTorch` • `Gymnasium` • `NumPy`

RNA Pairing Pattern Recognition Model — Mar 2024 – Jul 2024

- **Developed** a system to predict RNA secondary structures using SCFGs, evolutionary information, and enhanced CYK parsing for complex motifs; presented at CBRC Journal Club.
- *Stack:* `NetworkX` • `Biopython` • `PhyML` • `NumPy/SciPy`

Skills

Programming Languages	`Python` • `JavaScript` • `TypeScript` • `R` • `Java` • `C++` • `Swift`
Frameworks	`React` • `Node.js` • `Express.js` • `Django` • `Electron`
Data Science / ML	`PyTorch` • `TensorFlow` • `Pandas` • `Scikit-learn` • `NumPy` • `OpenCV` • `LangChain`
Platforms & Tools	`Docker` • `Git` • `GitHub` • `MongoDB` • `PostgreSQL` • `Hugging Face`

Languages

Persian | Native — **English** | Fluent

Certifications

Bioinformatics Internship Program | Biocan — Nov 2025

Scientific Secretary: Dr. Z. Salehi • Course Director: Dr. K. Kavousi

Introduction to Bioinformatics | Biocan — Jul 2025

Scientific Chair: Dr. K. Kavousi • Score: 96/100 • Top 3% of 350+ students

Scrum Foundations Course | Ultima Training Tech — Dec 2024

Instructor: Josef Balahan

Volunteering

Student Scientific Magazine "Halghe" | Amirkabir University of Technology — Dec 2025 – Present

Contributed to scientific editing and supported the editorial team

Technical & Web Support Volunteer | Amirkabir University of Technology — May 2023, Feb 2025

Web and procurement support for DAI DAY (2025) and the International Conference on AI and Smart Vehicles (2023)

References

Prof. Mehdi Ghatee | Scholar
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Professor of Electrical and Computer Engineering
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